

Weekly Running Sheet

Week Ending 7th Nov 2021

Cust ID	Customer Name	Cust Business Name	Function Room ID	Function Room Name	Booking Start Date Time	Booking End Date Time	Number Guests	Base Rate \$/Guest	Room Rate \$/Hour	Cuisine Type	Cuisine Rate \$/Guest	Total Cost	Decoration Theme
161	Margaret Ancoro	Monash University	25	Caulfield Sports Venue Hall	02/11/2021 18:00	02/11/2021 22:30	60	40	150	Chinese	30	\$4,875	
12	Wendy Natal				04/11/2021 18:00	04/11/2021 22:00	35	65		Western	25	\$3,150	Birthday
251	Pierre Wang		27	The Tudor Room	06/11/2021 19:00	07/11/2021 01:00	30	42	80	Chinese	30	\$2,640	Outdoor
309	George Sato		25	Caulfield Sports Venue Hall	06/11/2021 19:30	06/11/2021 21:30	70	35	150	Japanese	45	\$5,900	Outdoor
102	Glen Eira	Caulfield City Council	12	Bay Watch Venue	07/11/2021 18:00	07/11/2021 20:00	56	55	100			\$3,280	For All Occasions
456	Muriel Acharya				07/11/2021 19:00	08/11/2021 01:00	50	60		Indian	30	\$4,500	Wedding

UNF

Booking({Cust_ID, Cust_Name}, {Room_ID, Room_Name, Room_Rate}, Book_Start, Book_Finish, Book_Guests, Book_BRate, {Cuisine_Type, Cuisine_Rate}, Book_Cost, Book_Theme)

Repeating groups:

Cust_ID, Cust_Name

Room_ID, Room_Name, Room_Rate

Cuisine_Type, Cuisine_Rate

*Note although Book_Theme is technically a repeating group it cannot be meaningfully extracted without using a surrogate key. As the brief specifies surrogate keys cannot be used this has been left within the Booking entity.

1NF – Remove repeating groups, make atomic, identify keys

Booking(Cust_ID, Room_ID, Book_SDate, Book_STime, Book_FDate, Book_FTime, Book_Guests, Book_BRate, Book_Cost, Book_Theme, Cuis_Type)

Customer(Cust_ID, Cust_LName, Cust_FName)

Room(Room_ID, Room_Name, Room_Rate)

Cuisine(Cuis_Type, Cuis_Rate)

Partial dependencies:

None

*The brief specifies not to add attributes, however not breaking up the attribute 'Book_Start', 'Book_Finish' and 'Cust_Name' would violate the rule of atomicity, so these have been broken down to their constituent parts.

2NF – Remove partial dependencies

Booking(Cust_ID, Room_ID, Book_SDate, Book_STime, Book_FDate, Book_FTime, Book_Guests, Book_BRate, Book_Cost, Book_Theme, Cuis_Type)

Customer(Cust_ID, Cust_LName, Cust_FName)

Room(Room_ID, Room_Name, Room_Rate)

Cuisine(Cuis_Type, Cuis_Rate)

Transitive dependencies:

None

3NF – remove transitive dependencies

Booking(Cust_ID, Room_ID, Book_SDate, Book_STime, Book_FDate, Book_FTime, Book_Guests, Book_BRate, Book_Cost, Book_Theme, Cuis_Type)

Customer(Cust_ID, Cust_LName, Cust_FName)

Room(Room_ID, Room_Name, Room_Rate)

Cuisine(Cuis_Type, Cuis_Rate)

Full dependencies:

Cust_ID, Room_ID, Book_SDate, Book_STime -> Book_FDate, Book_FTime, Book_Guests,
Book_BRate, Book_Cost, Book_Theme

Cust_ID -> Cust_LName, Cust_FName

Room_ID -> Room_Name, Room_Rate

Cuis_Type -> Cuis_Rate

Sample Document 1:

Function Room Evaluation

Review Chef ID 12 Date Checked 11th Jun 2021
Name Matt Gilmore
Function Room ID 25
Name Caulfield Sports Venue Hall
Address 123 Wide Rd Caulfield
Capacity 80
On Site Parking Yes ☒ No ☐
Approved for use Yes ☐ No ☒
Optional Comments Several stoves in the kitchen are not working (max 200 chars) correctly, dishwasher non functional

Sample Document 2:

Function Room Evaluation

Review Chef ID 8 Date Checked 21st Sep 2021
Name Poh Ling Kwong
Function Room ID 27
Name The Tudor Room
Address 55 Narrow Lane Caulfield
Capacity 40
On Site Parking Yes ☐ No ☒
Approved for use Yes ☒ No ☐
Optional Comments (max 200 chars)

ONF

Evaluation(Eval_Date, {Chef_ID, Chef_Name}, {Room_ID, Room_Name, Room_Address, Room_Capacity, Room_Parking}, Eval_Result, Eval_Comments)

Repeating groups:

Chef_ID, Chef_Name

Room_ID, Room_Name, Room_Address, Room_Capacity, Room_Parking

1NF

Evaluation(Eval_Date, Chef_ID, Room_ID, Eval_Result, Eval_Comments)

Chef(Chef_ID, Chef_LName, Chef_FName)

Room(Room_ID, Room_Name, Room_StreetNum, Room_StreetName, Room_Suburb, Room_Capacity, Room_Parking)

Partial dependencies:

None

2NF

Evaluation(Eval_Date, Chef_ID, Room_ID, Eval_Result, Eval_Comments)

Chef(Chef_ID, Chef_LName, Chef_FName)

Room(Room_ID, Room_Name, Room_StreetNum, Room_StreetName, Room_Suburb, Room_Capacity, Room_Parking)

Transitive dependencies:

None

3NF

Evaluation(Eval_Date, Chef_ID, Room_ID, Eval_Result, Eval_Comments)

Chef(**Chef_ID**, Chef_LName, Chef_FName)

Room(**Room_ID**, Room_Name, Room_StreetNum, Room_StreetName, Room_Suburb,
Room_Capacity, Room_Parking)

Full dependencies:

Eval_Date, Chef_ID, Room_ID -> Eval_Result, Eval_Comments

Chef_ID -> Chef_LName, Chef_FName

Room_ID -> Room_Name, Room_StreetNum, Room_StreetName, Room_Suburb, Room_Capacity,
Room_Parking

Consolidation and attribute synthesis

1. Booking(**Cust_ID**, **Room_ID**, **Book_SDate**, **Book_STime**, Book_FDate, Book_FTime,
Book_Guests, Book_BRate, Book_Cost, Book_Theme, Cuis_Type)
2. Customer(**Cust_ID**, Cust_LName, Cust_FName)
3. Room(**Room_ID**, Room_Name, Room_Rate)
4. Cuisine(**Cuis_Type**, Cuis_Rate)
5. Evaluation(**Eval_Date**, **Chef_ID**, **Room_ID**, Eval_Result, Eval_Comments)
6. Chef(**Chef_ID**, Chef_LName, Chef_FName)
7. Room(**Room_ID**, Room_Name, Room_StreetNum, Room_StreetName, Room_Suburb,
Room_Capacity, Room_Parking)

3&7 can be combined;

Room(**Room_ID**, Room_Name, Room_StreetNum, Room_StreetName, Room_Suburb,
Room_Capacity, Room_Parking, Room_Rate)

Final consolidated result:

Booking(Cust_ID, **Room_ID**, **Book_SDate**, **Book_STime**, Book_FDate, Book_FTime, Book_Guests,
Book_BRate, Book_Cost, Book_Theme, Cuis_Type)

Customer(**Cust_ID**, Cust_LName, Cust_FName)

Room(**Room_ID**, Room_Name, Room_StreetNum, Room_StreetName, Room_Suburb,
Room_Capacity, Room_Parking, Room_Rate)

Cuisine(**Cuis_Type**, Cuis_Rate)

Evaluation(**Eval_Date**, **Chef_ID**, **Room_ID**, Eval_Result, Eval_Comments)

Chef(**Chef_ID**, Chef_Name)